🟢 PHASE 1: USER ONBOARDING & ACCOUNT CREATION

✅ 1. Landing Page

- ☐ Clear app purpose explanation
- ☐ “Sign Up” and “Login” buttons
- ☐ Parent-focused trust messaging
- ☐ Privacy policy & terms link

✅ 2. Parent Registration (Primary Account Holder)

- ☐ Parent Name
- ☐ Parent Email (Verified via OTP)
- ☐ Parent Mobile Number (Verified via OTP)
- ☐ Password creation
- ☐ Accept terms checkbox
- ☐ Email verification system
- ☐ Mobile OTP verification system

✅ 3. Child Profile Creation (Under Parent Account)

- ☐ Child Name
- ☐ Age
- ☐ Grade/Class
- ☐ School name (optional)
- ☐ Device type used (Laptop / Desktop / Mobile)
- ☐ Allow camera permission prompt
- ☐ Allow microphone permission prompt
- ☐ Allow location access permission
- ☐ Store device ID

✅ 4. Dashboard Redirect After Signup

- ☐ Redirect to Child Study Dashboard
- ☐ Show:
  - Child Name
  - Parent Name
  - Today's Study Summary
  - "Start New Study Session" button

🟢 PHASE 2: TASK SETUP BEFORE STUDY SESSION

✅ 5. Task Selection Screen

Task Input Options

- ☐ Parent predefined tasks list
- ☐ Child can add custom task
- ☐ Subject selection dropdown
- ☐ Estimated time input field
- ☐ Priority level option
- ☐ Multiple tasks add feature
- ☐ Edit/delete task option

✅ 6. Timer Setup

- ☐ Manual time selection (HH:MM)
- ☐ Quick presets (25 min / 45 min / 1 hr)
- ☐ Start Button
- ☐ Confirmation popup:
  - "Are you ready to focus?"

🟢 PHASE 3: LIVE STUDY SESSION (CORE ENGINE)

🔴 **A. Live Monitoring Engine**

✅ **7. Camera-Based Monitoring**

- ☐ Face detection enabled
- ☐ Detect if:
  - Face missing for 10–30 seconds
  - Looking away repeatedly
- ☐ If missing → Trigger Alert 1
- ☐ Store alert in database with timestamp

✅ **8. Mobile Distraction Detection**

- ☐ Detect phone usage via:
  - Head pose detection
  - Hand detection
- ☐ If phone detected → Trigger Alert 2
- ☐ Play alarm sound
- ☐ Log event in database

✅ **9. Tab Switching Detection (Web App)**

- ☐ Detect tab change event
- ☐ Detect minimized window
- ☐ Detect navigation to another URL
- ☐ Trigger Alert 3
- ☐ Play warning sound
- ☐ Log timestamp + URL attempted

🔴 **B. Alert System Logic**

✅ **10. Alert System Rules**

- ☐ If 1 alert → Show warning popup
- ☐ If 3 alerts → Deduct 1 Life
- ☐ Reset alert count after life deduction
- ☐ Store:
    - Alert type
    - Time
    - Duration
    - Session ID

🔴 **C. Gamification System**

✅ **11. Life System**

- ☐ Initial lives = 5
- ☐ Display life icons visually
- ☐ Deduct life after 3 alerts
- ☐ Show animation when life lost
- ☐ If lives = 0:
    - End session automatically
    - Trigger guardian alert

🔴 **D. Parent Alert System**

✅ **12. If All Lives Lost**

- ☐ Send Email to parent
- ☐ Send Mobile push notification
- ☐ Include:
    - Session duration
    - Total alerts
    - Live device location (Latitude/Longitude)
    - Google Maps clickable link

✅ 13. Session Completion Notification

- ☐ Send email on successful completion
- ☐ Include:
  - Task completed
  - Time spent
  - Location at completion
  - Alert summary

🔴 E. Psychological Exit Trap

✅ 14. Exit Prevention Logic

- ☐ If user tries closing tab:
  - Show popup:
    "Leaving means lying to yourself."
- ☐ Require typing "DONE" manually
- ☐ Prevent accidental exit
- ☐ Log forced exit attempts

🔴 F. Live Study Screen UI Must Show:

✅ 15. Live Dashboard Components

- ☐ Countdown Timer
- ☐ Task list
- ☐ Remaining lives
- ☐ Alert count
- ☐ Webcam live feed window
- ☐ AI Chatbot section
- ☐ Session progress bar

🟢 PHASE 4: BUILT-IN AI DOUBT SOLVER (Anti-Distraction System)

✅ 16. Integrated Chatbot Panel

- ☐ Text input box
- ☐ “Ask Doubt” button
- ☐ Connect to Gemini API
- ☐ Show answer inside panel
- ☐ Disable external links
- ☐ Keep user on same screen
- ☐ Log chatbot usage time

🟢 PHASE 5: SESSION END LOGIC

✅ 17. Timer Completion

- ☐ Play completion sound
- ☐ Show congratulation animation
- ☐ Award badge or points
- ☐ Store:
    - Total time
    - Total alerts
    - Lives left
    - Chatbot usage

🟢 PHASE 6: DAILY ANALYTICS DASHBOARD

✅ 18. Student Dashboard

Show:

- ☐ Total study time today
- ☐ Alert breakdown by type
- ☐ Lives lost

- ☐ Focus score percentage

✅ 19. Line Chart Report

- ☐ X-axis: Time
- ☐ Y-axis: Alerts
- ☐ Different color lines for:
  - Phone distraction
  - Face missing
  - Tab switching
- ☐ Zoom option
- ☐ Download PDF report

✅ 20. Parent Dashboard

- ☐ Weekly focus trend graph
- ☐ Daily comparison
- ☐ Most common distraction type
- ☐ Location logs history
- ☐ Email report auto-send option

🟢 PHASE 7: LOCATION TRACKING SYSTEM

✅ 21. During Session

- ☐ Capture location every X minutes
- ☐ Store coordinates securely
- ☐ Encrypt location data

✅ 22. On Completion / Failure

- ☐ Send final captured location

- ☐ Attach map preview link

🟢 **PHASE 8: DATABASE STRUCTURE**

✅ **23. Required Tables**

- ☐ Parents
- ☐ Children
- ☐ Sessions
- ☐ Alerts
- ☐ Location logs
- ☐ Chatbot logs
- ☐ Life deductions
- ☐ Tasks

🟢 **PHASE 9: SECURITY & PRIVACY**

✅ **24. Security**

- ☐ Encrypt passwords
- ☐ Encrypt location
- ☐ Secure camera stream
- ☐ HTTPS enabled
- ☐ JWT authentication
- ☐ Role-based access (Parent/Child)

🟢 **PHASE 10: MOBILE APP (Future – Same Flow)**

✅ **25. Replicate Features for Mobile**

- ☐ Face monitoring via front camera
- ☐ App switching detection
- ☐ Foreground lock system

- ☐ Push notifications
- ☐ Background location permission
- ☐ Same gamification model

🟢 **BONUS ADDITIONS (Highly Recommended)**

- ☐ Reward system (coins, badges)
- ☐ Leaderboard (optional)
- ☐ Pomodoro mode
- ☐ Dark mode
- ☐ AI study suggestions
- ☐ Focus music built-in

🧠 **CORE SYSTEM LOGIC SUMMARY FLOW**

1. Parent registers
2. Child profile created
3. Task added
4. Timer set
5. Live monitoring starts
6. Alerts tracked
7. Lives deducted
8. Parent notified if failure
9. Completion report generated
10. Analytics updated